

Post-Quantum Cryptography Migration

Part 19 - Finding & Governing It: Governance, Mandates & Timelines

Compiled by the Spaced Research Ledger

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Every earlier part of this collection describes technology. This one describes the deadlines. Governments have converted post-quantum guidance into dates with consequences, and the dates now drive vendor roadmaps more than the physics does.

Six sections cover the governance stack. Section 1 covers the U.S. federal mandates, Section 2 the European and British programs, and Section 3 the sector compliance regimes. Section 4 covers the certification schemes, Section 5 the operating-system policy switches that enforce it all, and Section 6 the procurement language reaching into contracts.

The year's pivot was June 22, 2026. Executive Order 14412 and the OMB memo that followed set a five-phase federal migration through 2035. Agency plans came due in months, with high-value assets on post-quantum key establishment by the end of 2030. France set the hardest commercial date anywhere: from 2027 its certification body stops certifying security products without quantum-resistant encryption, a market-access gate for critical infrastructure.

The uncomfortable finding sits between jurisdictions. The U.S. suite prefers pure post-quantum, France mandates hybrids, and Germany recommends algorithms NIST did not standardize. No single implementation satisfies all three regimes, and vendors now build for divergence.

1. US Federal Mandates

American policy runs on two tracks: national-security systems under NSA's CNSA 2.0 suite, and civilian agencies under White House memoranda now backed by statute. Both tracks got their operative documents in 2026. This section covers the chain of authority from memoranda to deadlines.

Recent Developments

The operative guidance finally landed. OMB Memorandum M-26-15, published two days after the June executive order, functionally supersedes the 2022 migration memo [1]. It establishes a five-phase 2026-to-2035 timeline and requires agency migration plans within 120 days [1]. It covers algorithm selection, hybrids, agility, and zero-trust integration, with the old inventory work folded into phase one [1].

It resolved an awkward gap: whether the statutorily required guidance had ever been issued by its 2025 deadline was publicly unclear [2]. On the national-security track, the IETF profiles are being finalized, with a CNSA 2.0 TLS profile using ML-DSA-87 and ML-KEM-1024 [3] and a certificate profile in final review [4].

Current State

The chain of authority is now complete. National Security Memorandum 8 directed the quantum-resistant update that became CNSA 2.0, with new national-security acquisitions compliant by January 2027 [5]. Its civilian twin directed annual inventories of quantum-vulnerable high-value assets and migration toward 2035 [5]. The 2022 OMB memo operationalized that, tying migration to zero trust, requiring inventories of classical key exchange, and, strikingly, prohibiting agency use of QKD for security functions without exception [6].

Its nine per-system data items proved mostly unautomatable, with only three realistically collectable by tools [7]. The Preparedness Act then elevated the whole structure to statute [5].

2. EU & UK Guidance

Europe governs by roadmap and Britain by phases, and both have now attached dates that bind. France's certification cutoff is the sharpest instrument in either jurisdiction. This section covers the continental machinery, the German technicalities, and the British capacity problem.

Recent Developments

France set the market gate. ANSSI will stop certifying security products lacking quantum-resistant encryption from 2027, expecting quantum-safe-only purchasing by 2030 [8]. The standards factories are producing. ETSI opened public enquiry on seventeen draft product standards under the Cyber Resilience Act [9], and its quantum-safe work program carries active drafts on signatures, PKI, and enterprise transport [10].

A second European migration conference is set for March 2027 [11]. Britain's NCSC published its industry workshop report, stressing that no organization migrates alone [12], and launched a consultancy pilot for discovery and migration planning through March 2027 [13].

Current State

The EU built its program on law. The 2024 Commission Recommendation grounds the coordinated roadmap in treaty authority, linked explicitly to the network-security directive [14]. The roadmap's milestones: member states begin by end-2026, critical infrastructure transitions no later than end-2030 [15], with full system-wide transition placed at 2035 [16]. A proposed directive would write post-quantum policy directly into national cybersecurity strategies [17], and a Quantum Act is expected in 2026 [18]. The Commission's technology chief framed the roadmap as the entry into the quantum era [15].

The technical posture is hybrid-by-default and risk-averse, preferring adjusted symmetric parameters and paired schemes [16], with practitioners noting comparable transitions have historically taken a decade [19]. Germany's technical guideline supplies the dates underneath: classical key exchange recommended only through 2031, hybrids mandatory from 2032, and classical signatures alone only through 2035 [20].

Britain published the cleanest phases. Identification and planning by 2028, highest-priority migrations by 2031, everything by 2035 [21], presented as the roadmap organizations asked for [21]. The annual review says the timelines now anchor plans across government and regulated sectors, while candidly admitting the

country lacks enough qualified consultancies [22]. A secondary characterization calls the NCSC the first major regulator to endorse the NIST algorithms, unverified [23].

3. Sector Compliance Regimes

Between government mandates and vendor products sit the compliance regimes that actually get audited: cloud authorization and payment-card rules. Neither yet requires post-quantum, and both now require knowing your cryptography. This section is short because the regimes are early.

Recent Developments

FedRAMP moved to visibility first. Its 2026 consolidated rules require cloud providers to document the cryptographic modules protecting federal data and their validation status [24]. Validated modules are encouraged rather than strictly required at moderate impact [25]. The OMB memo layers responsibility on top, requiring agencies to delineate migration duties with their cloud providers, with CISA and the defense department leading for shared services [26]. Analysts read the rules as compliance-plus-readiness scaffolding, not yet a post-quantum mandate [27].

Current State

The payment-card standard got there by accident of timing. PCI DSS 4.0's future-dated controls became fully mandatory in March 2025 [28]. Its required cryptographic inventory and deprecated-algorithm migration plan is the post-quantum planning artifact by another name [28].

4. Certification Schemes

Certification is where governance touches silicon: Common Criteria for chips, FIPS 140-3 for cryptographic modules. The first post-quantum certificates now exist on both tracks, and the schemes' calendars are forcing functions of their own. This section covers the firsts, the queue, and the divergence problem.

Recent Developments

The first certificates arrived in a cluster. France issued its first post-quantum Common Criteria certificates to Thales and Samsung in late 2025 [29]. A high-assurance certificate for a composite product on a post-quantum smart-card platform followed [30]. Germany certified the world's first smart card implementing a post-quantum algorithm [31].

On the FIPS track, Crypto4A's module achieved the first Level 3 validation for a quantum-safe HSM with the full NIST suite [32]. Sansec's followed as the first Chinese module [33]. The calendar pressure is structural. FIPS 140-2 modules serve new systems only through September 21, 2026 [34], and the implementation guidance added post-quantum self-test requirements [35].

The in-process list carries multiple post-quantum modules [36], and Europe's certification cryptography guidance went to public review with hybridization recommendations [37].

Current State

France's gate has teeth and history. The 2027 announcement came at a national conference from a senior official [38], and certification is required for French government and critical-infrastructure deployment, making it market access, not advice [39]. The signaling ran for years through position papers and qualification FAQs before the public date [40]. The commissioned survey behind it found no surveyed organization with a transition plan [40], and commentary places it in a wider trend of guidance hardening into 2028-to-2030 obligations [40].

The divergence is documented and unresolved. The U.S. suite prefers pure post-quantum, France mandates hybrids, and Germany mandates hybrids while recommending FrodoKEM and Classic McEliece over the NIST lattice choice, so no single implementation satisfies all three [40]. The chip certifications accumulate meanwhile: Infineon at the highest assurance level from Germany, Samsung from the Netherlands [41], three post-quantum HSM certifications across the two schemes [42].

The bottleneck is paperwork as much as product, with one TPM vendor sampling hardware it cannot yet certify because the post-quantum protection profile awaits approval [43]. On the FIPS side, a validated post-quantum wolfCrypt module sits in submission [44].

5. System-Wide Crypto Policy Enforcement

Policy only matters where an operating system enforces it. Linux distributions, Windows, Java, and Apple's platforms each expose the switches that turn post-quantum on, off, or mandatory across a whole estate. This section is the administrator's map of where those switches live and what they actually do.

Recent Developments

The defaults hardened. RHEL 10.1 enables post-quantum algorithms in every predefined policy, with a subpolicy to opt out [45]. RHEL 10's FUTURE policy now refuses classical key exchange entirely, breaking connections to anything unready [46]. Windows shipped its hybrid groups into production Schannel in July, three configurable options, disabled by default, TLS 1.3 only [47][47], documented in the platform's supported-groups reference [48]. Java's hybrid exchange arrives with JDK 27 and backports through 2027 [49], and Apple documented a diagnostic command for verifying quantum-secure connections [50].

Current State

RHEL is the most complete policy machine. Four predefined levels govern the system through one tool [51], with a post-quantum subpolicy layerable on any base since RHEL 9.7 [52]. A legacy level trades security for compatibility [52], and services need restarts after changes [51]. The honest caveat: applications can override, making policy a default rather than an absolute boundary [53].

The web console exposes it graphically [54]. The current state of the defaults: every predefined level prefers post-quantum in 10.1 [55], and FIPS mode gains hybrid SSH exchanges in 10.2 [56]. Red Hat turned hybrids on for its own update infrastructure [57]. FIPS mode itself tightened.

The setup tool goes beyond the policy switch [58], and enforcement hinges on the kernel flag [58]. RHEL 10 removed post-install FIPS conversion entirely, install-time or reinstall [59], following long-standing advice to enable it from the start [60].

Windows and Java put the switches where administrators already look. The hybrid groups configure through the existing curve-order Group Policy, device management, or PowerShell [61], deliberately requiring explicit action [62]. Fallback to classical is silent, with event-log confirmation of what actually negotiated [63]. Certificate issuance is a separate track, generally available since May, leaving existing hierarchies untouched [47].

Java's control plane is its security-properties file, disabling algorithms JVM-wide [64], inspectable at runtime [64]. The file moved location in JDK 9 [65], and algorithms re-enable by edit or API before the TLS subsystem initializes [66]. It is a security property, not a command-line flag [67], one of a family covering certificates, JARs, and curves [66], joined in JDK 26 by an API-level disablement property [68].

Apple manages by profile and enforces by transport. Configuration profiles carry policy from device management, lockable on supervised devices [69], signed and CMS-encrypted [69]. The management channel's own TLS requirements are strict and blocking, from minimum protocol versions to key sizes [70], applying to enrollment, profiles, apps, and updates alike [70]. And the endpoint default is already set, with quantum-secure key exchange on by default in the 26-generation networking APIs [71].

6. Procurement, Contract Language & Audit

The last mile of governance is a purchase order. Executive orders flow into acquisition regulations, vendor questionnaires, and eventually insurance premiums, which is how mandates reach companies that never read a NIST publication. This section covers the contract language taking shape and the audit posture forming behind it.

Recent Developments

The executive order started the acquisition clock. It begins moving federal contractors toward a governmentwide post-quantum compliance regime [72]. It directs the FAR Council to propose vulnerability-disclosure rules covering cryptographic flaws within 270 days [73]. It also orders CISA and NIST to define the minimum elements of a cryptographic bill of materials for automated assessment [74]. The defense side moves in parallel, with the CMMC transition to the newer control revision offering a vehicle for post-quantum clauses ahead of rulemaking [75].

The questionnaires are becoming standard artifacts. Australia released vendor-neutral post-quantum questions for procurement and vendor assurance [76], spanning dependencies, planning, and supplier communication [77], recommending vendors maintain a current CBOM down to embedded components [76]. A Navy solicitation seeks prototypes automating discovery and CBOM generation [78], and CISA's quantum lead said agencies will write quantum-resistance into contract language [79]. The defense department requested industry input on encryption aligned to the migration timelines [80], and Treasury's new task force carries a dedicated vendor-readiness workstream [81]. One survey punctures the confidence: three-quarters

of enterprises claim continuously updated inventories while nearly half have no single owner for the migration [82].

Current State

The contractual baseline is shifting under the analysis. As of the general assessments, post-quantum is not yet a federal contractual requirement [83]. The defense certification regime still rests on the older control revision [83], and cloud authorization awaits control-catalog updates and validated implementations expected no earlier than 2027 [83]. The executive order's trajectory is explicit: proposed acquisition rules by December 2026, contractor compliance with the post-quantum FIPS standards by the end of 2030 [72][75], plus disclosure-program updates [84].

The practical near-term effect is inventory obligations and vulnerability assessment for systems behind high-value assets [85]. CISA is compiling an approved cryptographic product list agencies will buy from [86], with agency leadership directed to prioritize harvest-now-decrypt-later exposure [86].

The buyer-side toolkit is filling out. GSA's buyer guide makes agility, secure-by-design, and interoperability the acquisition criteria [86]. The PKI Consortium's maturity model gives procurement a six-level vendor yardstick [87]. Europe's directives convert vendor assurance into obligation, with supply-chain security required under one and third-party ICT risk under the other, neither satisfied by self-certification [88].

Canada recommends contract clauses requiring approved post-quantum algorithms by end-2026 [89], and the FAA's request for information asked vendors directly for algorithm and transition detail [90]. The audit backstop is arriving through insurance, with underwriters folding quantum readiness into renewals [83] and weak transition plans risking premiums or exclusions [83]. Beneath it all, the inventory remains the non-negotiable baseline every policy document starts from [83], as NIST's practice guide established with its industry collaborators [91][91].

Unresolved Conflicts

One agency got ahead of the analysis. The general claim that post-quantum is not yet a federal contractual requirement [83] collides with the USDA's acquisition regulation, revised September 2025 [83]. That regulation explicitly requires solicitations for CISA-listed product categories to demand post-quantum support [83]. The likeliest reading is an agency-specific exception to a true general rule, but the sources do not reconcile it. A governmentwide acquisition tracker would settle it.

About This Report

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